

COURSE: CBIO313 – Data Mining & Machine Learning

INSTRUCTOR: Dr. Muhammad Elsayeh

PROJECT TITLE: Diabetes Risk Prediction Model Using Machine Learning

STUDENT NAME: Malak Amr

STUDENT ID: 221001703

1. PROBLEM DEFINITION & OBJECTIVES

Diabetes mellitus is a chronic metabolic condition characterized by elevated blood glucose levels. Early diagnostic prediction plays a crucial role in preventing long-term cardiovascular, renal, and neurological complications. The primary objective of this project is to construct a machine learning pipeline to classify patients as diabetic or non-diabetic based on non-invasive clinical attributes, facilitating early medical intervention.

2. DATASET DESCRIPTION

The project utilizes the Pima Indians Diabetes Dataset sourced from the UCI Machine Learning Repository. The dataset contains 768 patient records with 8 numerical biological predictor variables and 1 binary outcome target:

- Pregnancies: Number of times pregnant
- Glucose: Plasma glucose concentration (2-hour oral glucose tolerance test)
- BloodPressure: Diastolic blood pressure (mmHg)
- SkinThickness: Triceps skinfold thickness (mm)
- Insulin: 2-Hour serum insulin (mu U/ml)
- BMI: Body mass index (weight in kg / (height in m)²)
- DiabetesPedigreeFunction: Genetic diabetes risk score
- Age: Age in years
- Outcome: Target class (0 = Non-diabetic, 1 = Diabetic)

3. DATA PREPROCESSING & EXPLORATORY DATA ANALYSIS (EDA)

- Data Cleaning & Imputation: Biological measurements where values of 0 are physiological impossibilities (Glucose, Blood Pressure, Skin Thickness, Insulin, BMI) were identified as missing data and replaced using median feature imputation.
- Feature Scaling: Applied standard normalization (StandardScaler) to ensure balanced feature weighting across distance-sensitive algorithms.
- Data Partitioning: Split the dataset into 80% training data and 20% testing data using stratified sampling to maintain uniform class proportions.
- Exploratory Data Analysis: Correlation heatmaps confirmed that Glucose concentration and BMI exhibit the strongest individual correlations with positive diabetes outcomes.

4. MACHINE LEARNING IMPLEMENTATION

Three supervised classification models were trained and evaluated:

1. Logistic Regression: A baseline linear classification algorithm.
2. Decision Tree Classifier: A rule-based non-linear tree model.
3. Random Forest Classifier: An ensemble model comprising 100 decision trees to improve generalization and mitigate overfitting.

5. PERFORMANCE EVALUATION & MODEL COMPARISON

Models were tested on unseen evaluation data using key statistical performance metrics:

- Logistic Regression: Accuracy = ~75%, Precision = ~67%, Recall = ~58%, F1-Score = ~62%
- Decision Tree: Accuracy = ~71%, Precision = ~59%, Recall = ~61%, F1-Score = ~60%
- Random Forest: Accuracy = ~78%, Precision = ~71%, Recall = ~65%, F1-Score = ~68%

Findings: The Random Forest Classifier achieved the highest overall predictive accuracy (~78%) and superior balance between precision and recall, proving to be the most effective algorithm for this dataset.

6. GRAPHICAL USER INTERFACE (BONUS IMPLEMENTATION)

To satisfy the project bonus requirements, an interactive graphical application was built using Streamlit (app.py). The app allows clinicians to enter raw patient biological parameters and receive instantaneous risk predictions powered by the trained Random Forest model.

7. CONCLUSION

The experimental results demonstrate that machine learning models—specifically Random Forest ensembles—can serve as reliable preliminary diagnostic decision support tools for diabetes detection. Future iterations can explore hyperparameter tuning, neural network architectures, and direct electronic health record (EHR) database integration.